

Randomized and low-rank approximation

Spectral-gap-independent constants for randomized block polynomial interpolation

Difficulty: challenging **Importance:** interesting to the community**Status:** [Lean verified](#)

Rating rationale: Challenging because almost-sure invertibility does not yield spectrum-uniform probabilistic constants; community impact is explaining cluster robustness of block Lanczos.

Negative resolution - 12 September 2026 (UTC)

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The answer to the displayed uniform-probability question is **no**. The [exact target](#) and [Sections 1-5 of the complete proof](#) give deterministic admissible blocks with $b = 2, d = 3$,

$$\Lambda_1 = 2I_2, \quad \Lambda_2 = \text{diag}(\varepsilon, 2\varepsilon), \quad \Lambda_3 = \text{diag}(0, 1).$$

Along $\varepsilon_m = 1/(m + 5)$, the probability that $\chi_{\text{mono}}\chi_{\text{coef}}$ is below any fixed finite constant tends to zero. This contradicts the requested bound already at $\delta = 1/2$. The proof uses the literal recurrence and all prescribed root orderings; no input depends on the sampled Gaussian matrices.

[Proof PDF](#) · [Standalone proof TeX](#) · [Independent mathematical review](#) · [Submission and public-source audit](#).

The complete proof passed a separate Codex-agent informal audit. Substantial AI assistance is disclosed; this is neither external human peer review nor formal verification. Shao retains credit for the original conjecture and interpolation framework. The counterexample uses interlaced block spectra and a repeated eigenvalue within one block, both permitted by the original hypotheses. It does not address a different question imposing ordered spectral intervals or refute cluster-robust block Lanczos convergence itself. The original statement, permanent ID, references and dated history remain below; the ratings are historical.

Lean verification — 15 September 2026

The complete original uniform-probability conjecture has a formally verified negative answer. The proof retains the literal descending recurrence, all root orderings, actual Euclidean operator and coefficient norms, and the full independent Gaussian law. A deterministic two-by-two, three-block family makes the actual interpolation constant diverge almost surely along a fixed sequence, and its probability of lying below any finite constant tends to zero. All ten reviewed statements passed LeanCert kernel trust audits, standard-axiom checks and [fresh isolated Linux Comparator/default-kernel verification](#) at the [immutable proof revision](#).

Mathematical proof: George Stepaniants. Original interpolation framework and conjecture: Nian Shao. Formalization: Sidney Holden, with OpenAI Codex assistance. Two independent statement reviews preceded implementation, and two independent nonauthor final reviews passed. Actual sandbox and rejection controls passed. AI-agent review is not external human peer review or official Tau Ceti endorsement.

[Proof and reviews](#) · [Retained evidence](#).

Original statement (retained)

Fix integers $b \geq 1, d \geq 2$. Let $\Lambda_1, \dots, \Lambda_d \in \mathbb{R}^{b \times b}$ be diagonal with pairwise disjoint spectra. Draw all entries of $\Omega_1, \dots, \Omega_d \in \mathbb{R}^{b \times b}$ independently from $N(0, 1)$ and put $B_i = \Omega_i^{-1} \Lambda_i \Omega_i$. Let a and c be the smallest and largest diagonal entries in the entire family.

For each $k \in \{1, \dots, d\}$, reorder the triples $(B_i, \Lambda_i, \Omega_i)$ into the order $(k, 1, \dots, k-1, k+1, \dots, d)$ and use superscript (k) for that ordering. In descending order $i = d, d-1, \dots, 1$, compute

$$S_{i,i}^{(k)} = I_b, \quad S_{i,j}^{(k)} = B_i^{(k)} S_{i,j-1}^{(k)} - S_{i,j-1}^{(k)} \hat{B}_j^{(k)} \quad (j = i+1, \dots, d),$$

$$\hat{\Omega}_i^{(k)} = \Omega_i^{(k)} S_{i,d}^{(k)}, \quad \hat{B}_i^{(k)} = (\hat{\Omega}_i^{(k)})^{-1} \Lambda_i^{(k)} \hat{\Omega}_i^{(k)}.$$

All norms below are spectral norms. Define

$$\chi_{\text{mono}}^{(k)} = \max_{2 \leq i \leq d} \left\{ 1, \frac{\|aI - \hat{B}_i^{(k)}\|_2}{\|aI - \Lambda_i^{(k)}\|_2}, \frac{\|cI - \hat{B}_i^{(k)}\|_2}{\|cI - \Lambda_i^{(k)}\|_2} \right\},$$

$$\chi_{\text{coef}}^{(k)} = \|(S_{1,d}^{(k)})^{-1}\|_2^{1/(d-1)} \min_{\substack{2 \leq i \leq d, \lambda \in \sigma(\Lambda_i^{(k)}) \\ \eta \in \sigma(\Lambda_i^{(k)})}} |\lambda - \eta|, \quad \chi_{\text{mono}} = \max_k \chi_{\text{mono}}^{(k)}, \quad \chi_{\text{coef}} = \max_k \chi_{\text{coef}}^{(k)}.$$

The inverses exist almost surely, as shown in the source. If an endpoint ratio is $0/0$, set it to 1; its matrix is necessarily a scalar matrix and the surrounding maximum already includes 1.

Is the family of random variables $\chi_{\text{mono}} \chi_{\text{coef}}$ uniformly bounded in probability over all admissible diagonal data? Precisely, for each $0 < \delta < 1$, does there exist a finite $C(b, d, \delta)$ such that, for every fixed admissible $(\Lambda_1, \dots, \Lambda_d)$,

$$\Pr\{\chi_{\text{mono}} \chi_{\text{coef}} \leq C(b, d, \delta)\} \geq 1 - \delta?$$

The constant must be independent of the eigenvalues and all gaps within and between their blocks. Probability is taken separately for each fixed input, not for one draw required to work simultaneously for every spectrum. This quantifies the source's "with high probability" statement with its explicitly allowed failure-probability dependence.

These constants control matrix-polynomial interpolation in the convergence analysis of randomized small-block Lanczos. The bound would explain robustness to clusters much larger than the starting block. The scalar-block case is already bounded deterministically; noncommutativity is the unresolved obstacle.

References

N. Shao, *A structural bound for cluster robustness of randomized small-block Lanczos*, arXiv:2507.10144v2, 30 May 2026, Theorem 1, equation (12), and Conjecture 1 in §3.2 ([primary full text](#)). T. Chen et al., *Does block size matter in randomized block Krylov low-rank approximation?*, arXiv:2508.06486, §§4,6 ([primary paper](#)), is related: its conjecture changes spectral-gap dependence in output bounds, whereas this question bounds the intermediate random interpolation constants themselves.

Status check — 2026-09-08

Latest arXiv abstract/history and May 2026 v2 inspected. The revision explicitly retains Conjecture 1, including independence of gaps within and between blocks. Searches for author/title with “proof”, “conjecture”, “cluster robustness”, and 2026 found no resolution. Generic nonsingularity of the recurrence does not give the claimed uniform probabilistic bound.

Audit — 2026-09-10

Rechecked [Shao’s May 2026 Conjecture 1](#) and current version history. It still requires independence of gaps inside and between blocks. Cluster-robustness and later-author searches found no resolution. The scalar-block baseline does not resolve noncommuting block interpolation.

Resolution audit - 2026-09-12

The independent review checked the full algebra, almost-sure nonsingularity of every prescribed ordering, and the deterministic-input probability quantifiers. A fresh public fork/branch/PR audit and primary-source search found no prior resolution of KE-05 at the recorded time; its exact scope and limits are in the linked submission record. All original mathematical hypotheses remain unchanged.